

ANALYZING STRESS IN A **WORKPLACE**

INTRODUCTION

- Work stress is an increasingly prevalent issue with significant impacts not only to employees well-being but also an organization's performance.
- Understanding the factors that contribute to work stress is important so as to develop effective strategies to mitigate its negative effects.
- This analysis aims to investigate the employees well-being hence determining the organization's overall health

DATA OVERVIEW

- The dataset explores the connections between lifestyle choices, health and work-related factors in relation to the stress levels among employees.

The key variables analyzed are:

- **AGE** : Employee's age in years
- **OCCUPATION** : Employee's job title
- **PHYSICAL ACTIVITY** : Level of physical activity
- **HEALTH** : Blood Pressure, Blood Sugar levels and Cholesterol levels
- **WORK HOURS** : Hours spent working
- **STRESS DETECTION** : Self-reported stress levels

SIMPLIFIED OVERVIEW OF THE DATASET

Age	Occupation	Physical Activity	Blood Pressure	Blood Sugar Level	Cholesterol Level	Work Hours	Stress Detection
30	Software Engineer	2	120	90	180	8	Low
35	Marketing Manager	1	110	80	160	9	Medium
40	Data Scientist	2	130	100	200	8	High

RESEARCH QUESTIONS

- Does sleep duration, sleep quality, physical activity, screen time and social interactions correlate with stress levels?
- How does smoking habit, meditation practice and exercise type relate to stress detection?
- Does age influence employees health and how does it relate to stress?
- Are certain occupations associated with higher stress levels and how does it relate to work hours and travel time?

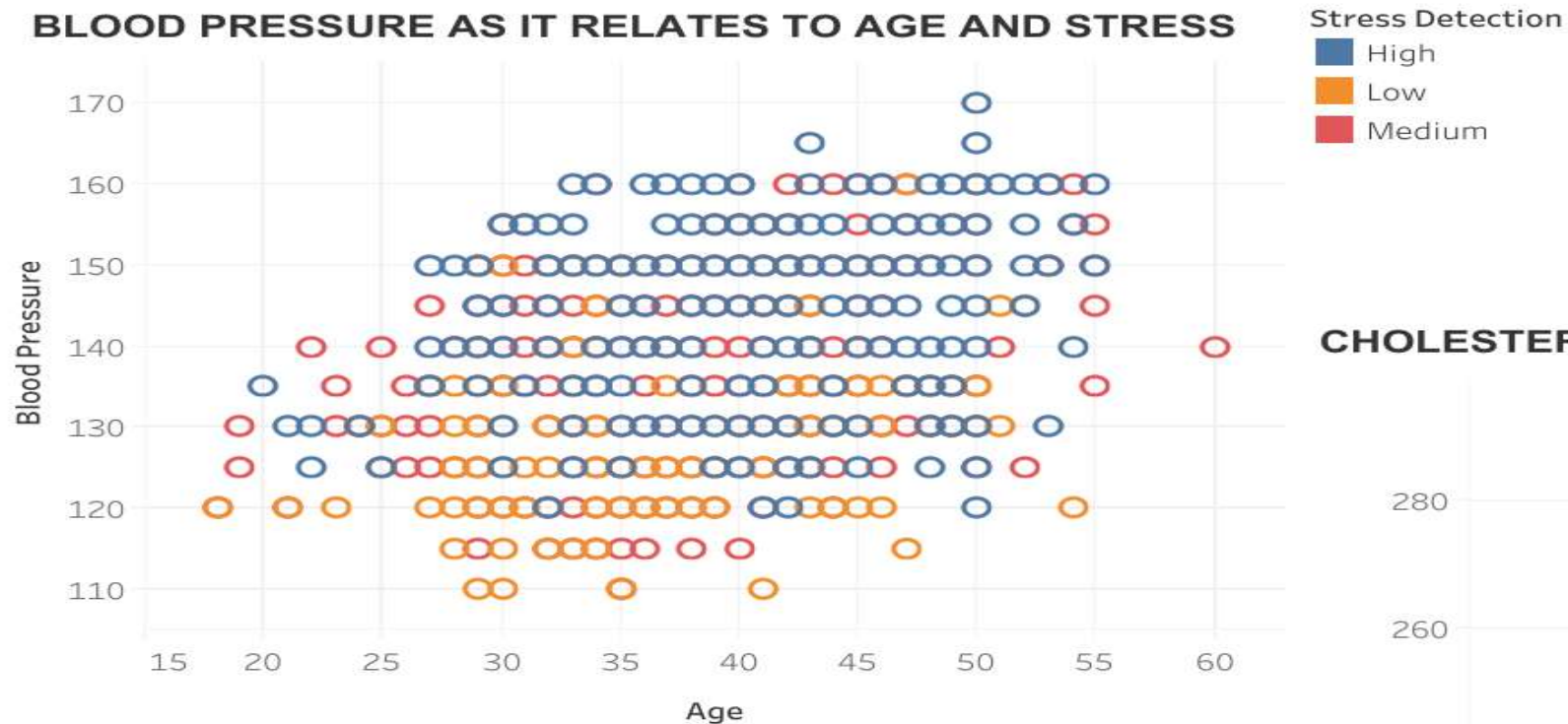
THE RELATIONSHIP BETWEEN SLEEP DURATION, SLEEP QUALITY AND STRESS



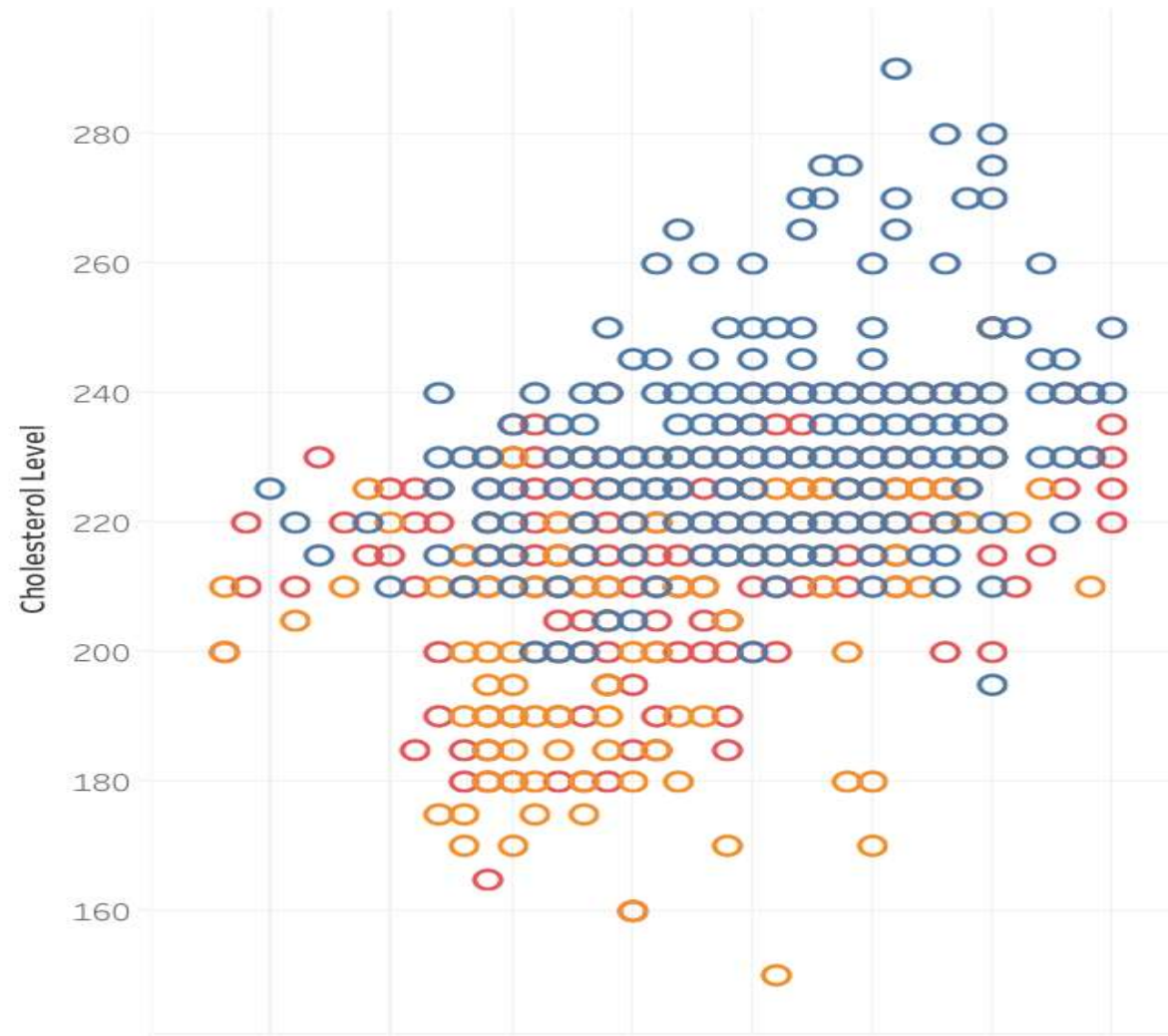
SLEEP DURATION, QUALITY AND STRESS ANALYSIS

- There is a positive correlation between sleep duration and sleep quality. The longer the sleep duration, the higher sleep quality.
- High stress is frequently associated with shorter sleep durations and lower sleep quality.

BLOOD PRESSURE AS IT RELATES TO AGE AND STRESS



CHOLESTEROL LEVEL AS IT RELATES TO AGE AND ST



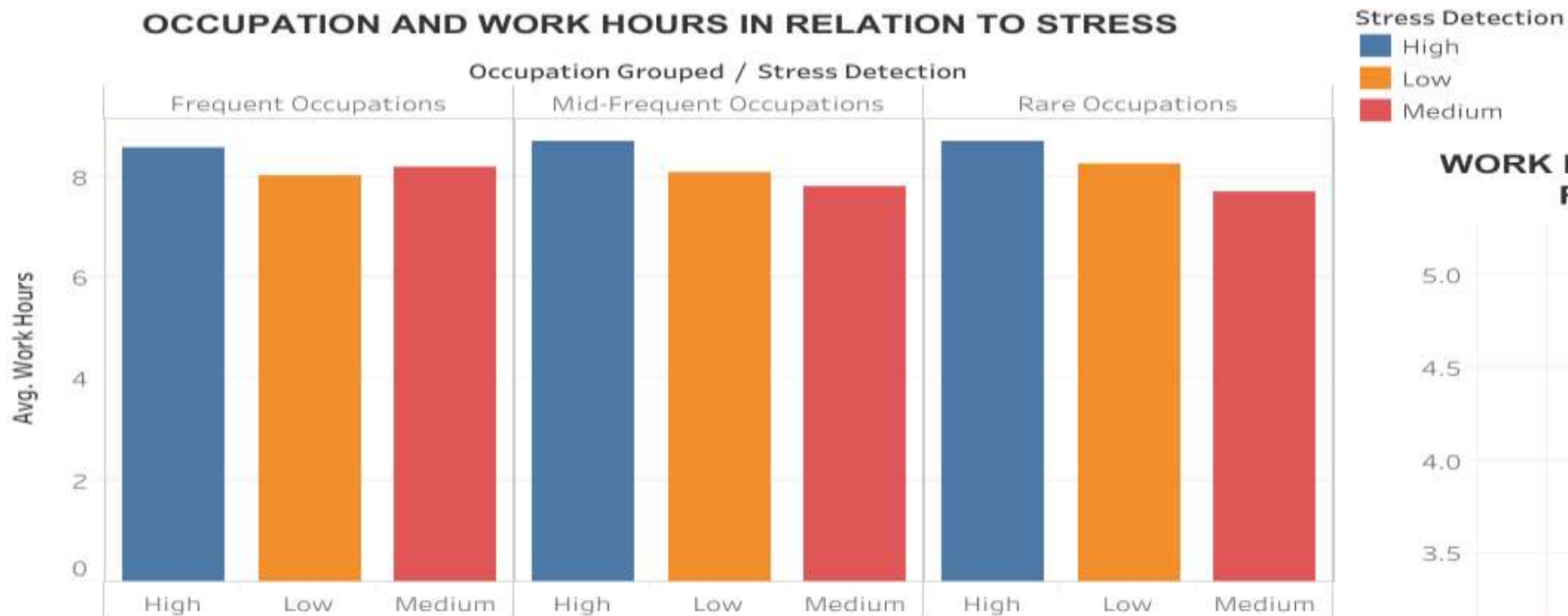
BLOOD SUGAR LEVEL AS IT RELATES TO AGE AND STRESS



AGE, HEALTH AND STRESS ANALYSIS

- There is a general trend of blood pressure, blood sugar levels and cholesterol levels increasing with age.
- High stress category has employees with the highest blood pressure, blood sugar levels and cholesterol levels and the impact of stress on all three might increase with age.

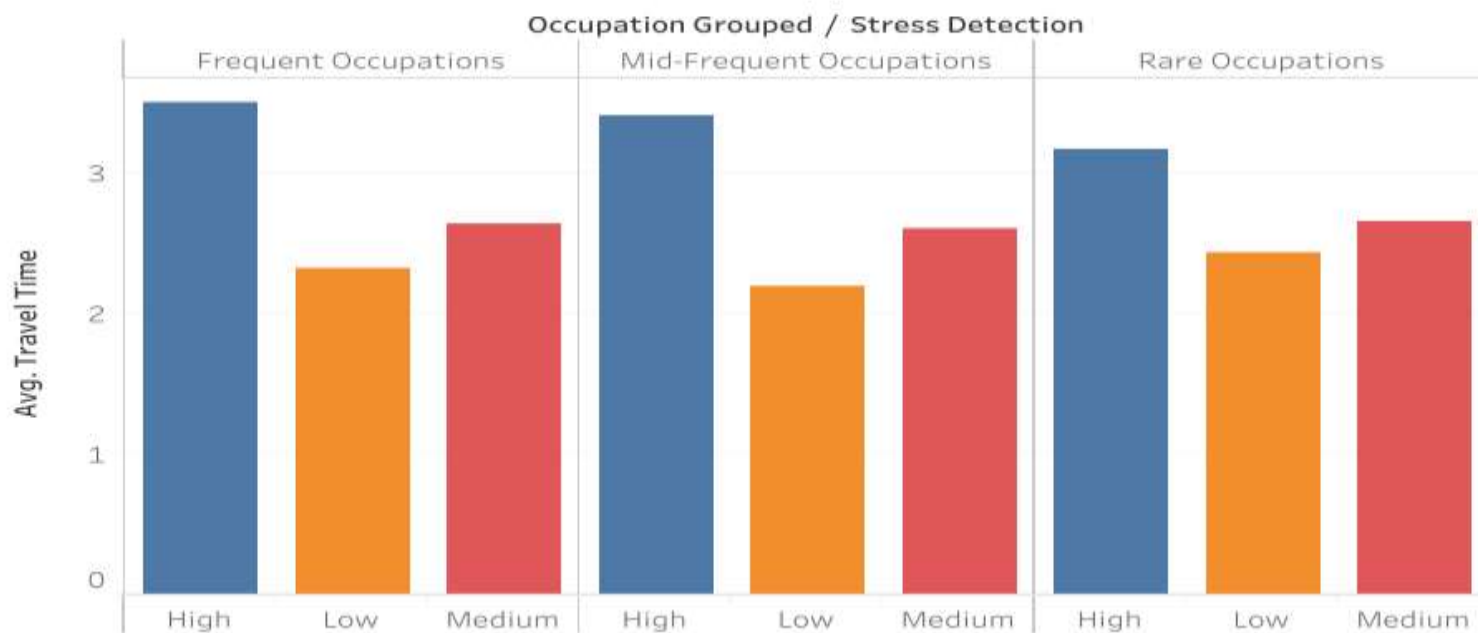
OCCUPATION AND WORK HOURS IN RELATION TO STRESS



WORK HOURS AND TRAVEL TIME IN RELATION TO STRESS



OCCUPATION AND TRAVEL TIME IN RELATION TO STRESS



WORK RELATED FACTORS AND STRESS ANALYSIS

- In all three occupation groups, employees reporting high stress tend to work longer hours on average and have long travel times.
- Employees with shorter work hours and travel time report having low stress.

RECOMMENDATIONS

- Flexible work schedules for employees so as to promote a work-life balance.
- Offer regular health screenings
- Implement wellness programs tailored for different age groups
- Investigate potential staffing shortage
- Make the workplace a smoke free environment

LIMITATIONS

- Dataset information is mainly self-reported hence subject to bias
- Correlation suggests a potential link of the observed relationship which is not definitive
- Measurement of stress

CONCLUSION

- Implementation plan starting with the easiest and one with the most impact
- Evaluate the effectiveness of the changes in the organization and assess if there are downsides.